

MCQs – Self Assessment

Chapter 29 (Magnetic Fields due to Currents) & Chapter 26 (Resistance & Resistivity)

1) At a distance R from a current carrying wire, what is the direction of the magnetic field relative to the wire?

- a) radially toward the wire
- b) radially away from the wire
- c) parallel to the wire
- d) in the direction opposite to that of the current
- e) in the direction that is perpendicular to both the wire and to the radial direction

2) A magnetic field is generated by a current-carrying wire. Which one of the following statements concerning this situation is false?

- a) The magnitude of this magnetic field decreases with increasing distance away from the wire.
- b) A right-hand rule is useful for determining the direction of the magnetic field at a particular location.
- c) The magnitude of the magnetic field is directly proportional to the magnitude of the current.
- d) The magnetic field is parallel to the direction of the current in the wire.

3) Consider two parallel wires carrying current in the same direction. Which one of the following statements to true concerning this situation?

- a) The two wires will attract each other, even if no external magnetic field is applied to the wires.
- b) The two wires will repel each other, even if no external magnetic field is applied to the wires.
- c) The two wires will attract each other, only if an external magnetic field is applied to the wires.
- d) The two wires will repel each other, only if an external magnetic field is applied to the wires.
- e) The wires will be neither attracted nor repelled from each other when no external magnetic field is applied to the wires.

4) Which one of the following parameters is not used to determine the magnetic force on a current-carrying wire in a magnetic field?

- a) length of the wire
- b) radius of the wire
- c) direction of the magnetic field with respect to the direction of the current
- d) the strength of the magnetic field
- e) the magnitude of the electric current

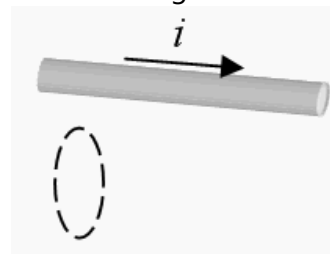
5) Ampere's law may be written as:

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{\text{enc}}$$

Consider the circular closed loop located near a current carrying wire as shown. What does the left side of the above equal for the closed loop if the current is directed to the right and has a magnitude of 2.0 A? The center of the loop, which has a radius of 2.5 cm, is located 4.0 cm from the wire.

- a) zero
- b) 3.1 T·m

- c) 4.8 T·m
- d) 7.2 T·m
- e) This cannot be determined with only the information given.



6) What is a solenoid?

- a) a single loop of wire in the shape of a circle
- b) a radio antenna
- c) a long coil of wire in the shape of a helix
- d) a scanning mechanism inside of a television
- e) a magnet that is inserted into a coil of wire

7) How is the direction of current flow defined in a conductor?

- a) It is in the direction of the force on a charged particle.
- b) It is in the direction that the atoms move.
- c) It is in the direction that positively-charged particles would move.
- d) It is in the direction that negatively-charged particles would move.
- e) It is up to the person doing a given problem to decide the direction.

8) Complete the following statement: Current density is a measure of

- a) the number of charges in a volume at a given time.
- b) the amount of current flowing through a volume.
- c) the amount of charges in a cross-sectional area at a given time.
- d) the amount of current flowing through a cross-sectional area.
- e) the total mass of the charges through a cross-sectional area.

9) Which pair of terms correctly fills the blanks in the following sentence? _____ is a property of an object while _____ is a property of a material.

- a) resistivity, conductivity
- b) current, current density
- c) current, resistance
- d) resistance, current
- e) resistance, resistivity

10) Which one of the following statements concerning Ohm's law is true?

- a) Ohm's law is true for all electronic devices.
- b) Ohm's law is true when the resistivity of a material is independent of the applied electric field.
- c) Ohm's law is true when the resistance varies linearly with the magnitude of the applied electric field.
- d) Ohm's law is true for all conductors.
- e) Ohm's law is true for all materials.